Capstone Two: Restaurant Rating Prediction & Insights

1. Problem Statement

The global restaurant industry is highly competitive, with ratings playing a critical role in customer decision-making and business success. The goal of this project is to identify key factors influencing restaurant ratings and to build a predictive model capable of estimating restaurant ratings based on measurable features. These insights can help restaurant owners and stakeholders improve performance and customer perception.

2. Data Overview

The dataset consists of top-rated restaurants worldwide and includes features related to:

- Price range
- Cuisine type
- Location
- Dining style
- Service characteristics
- Review-based attributes

After data cleaning and preprocessing, the final modeling dataset was split into training and testing sets to ensure unbiased model evaluation.

3. Data Wrangling & Preprocessing

Key steps included:

- Handling missing values
- Encoding categorical variables
- Feature scaling where appropriate
- Train-test split using preprocessed datasets

These steps ensured the data was suitable for machine learning while preserving interpretability.

4. Exploratory Data Analysis (EDA)

EDA revealed several meaningful trends:

- Restaurants with higher price ranges tend to have higher ratings.
- Certain cuisine categories consistently outperform others.
- Location plays a moderate role, suggesting regional preferences.
- Rating distributions are slightly skewed, indicating fewer low-rated establishments in the dataset.

Visualizations were used to validate assumptions and guide feature selection.

5. Modeling Approach

Multiple regression-based and ensemble models were tested. The final model was selected based on:

- Predictive accuracy
- Generalization performance on test data
- Interpretability

The best-performing model balanced accuracy and explainability, making it suitable for real-world business use.

6. Model Performance

The final model demonstrated:

- Strong predictive performance on unseen test data
- Low error variance
- Stable results across evaluation metrics

Predicted ratings closely aligned with actual values, confirming the model's reliability.

7. Key Findings

- Price range is the strongest predictor of restaurant rating.
- Cuisine type has a measurable impact on customer perception.
- Service-related features significantly influence overall ratings.
- Location matters, but less than pricing and service quality.

8. Business Recommendations

Based on the findings, restaurants can:

1. Optimize pricing strategy to match perceived quality and customer expectations.
2. Invest in service improvements, which strongly correlate with higher ratings.
3. Focus on cuisine differentiation, especially in competitive markets.

9. Future Work

Potential next steps include:

- Incorporating customer sentiment analysis from reviews
- Exploring time-based trends in ratings
- Testing deep learning models for nonlinear relationships
- Expanding the dataset to include lower-rated restaurants

10. Conclusion

This project successfully demonstrates that restaurant ratings can be predicted with meaningful accuracy using structured data. The insights gained provide actionable strategies for improving customer satisfaction and business outcomes.